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Numerical Methods

Unit 4: The Runge-Kutta 4th order method

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Outline

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The Runge-Kutta 4th order method

Originally developed by Carl Runge and Martin Kutta, this approach is formally known as the classic fourth-orde Runge-Kutta method, though it is frequently referred to simply as the **Runge-Kutta method**. The general formula is:

$$y_{i+1} = y_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)h,$$

where the intermediate slopes are calculated as:

- $k_1 = f(t_i, y_i)$
- $k_2 = f(t_i + \frac{1}{2}h, y_i + \frac{1}{2}k_1 h)$
- $k_3 = f(t_i + \frac{1}{2}h, y_i + \frac{1}{2}k_2 h)$
- $k_4 = f(t_i + h, y_i + k_3 h)$

Remarks

- The local (single) truncation error is proportional to h^5 and that for a finite interval the global truncation error is at most a constant times h^4.
- if f does not depend on y, then

$$y_{i+1} - y_i = \frac{h}{6}\left[f(t_i) + 4f(t_i + \frac{h}{2}) + f(t_i + h)\right],$$

This is **Simpson's rule** for the approximate evaluation of the integral of $y' = f(t)$.

Comparison of RK Orders

- **Euler's Method (1st Order):** Significant deviation from the exact solution.
- **Heun's Method (2nd Order):** Better tracking but still carries significant error.
- **RK 4th Order:** Closely aligns with the exact result.

Kutta's 4th Order Formula

Other versions of the 4th order method exist, such as Kutta's formula:

$$y_{i+1} = y_i + \frac{1}{8}(k_1 + 3k_2 + 3k_3 + k_4)h$$

- $k_1 = f(t_i, y_i)$
- $k_2 = f(t_i + \frac{1}{3}h, y_i + \frac{1}{3}hk_1)$
- $k_3 = f(t_i + \frac{2}{3}h, y_i - \frac{1}{3}hk_1 + hk_2)$
- $k_4 = f(t_i + h, y_i + hk_1 - hk_2 + hk_3)$

This is equivalent to **Simpson's 3/8 rule** if f is only a function of t.

Example

Use the classical Runge–Kutta method (RK4) with step sizes $h = 0.025, 0.01,$ and 0.001 to find approximate values of the solution of the following initial value problem the interval $0 \leq t \leq 2$.

$$y' = \frac{t}{y}, \qquad y(1) = 2.$$

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First Iteration ($h = 0.025$)

At $t_0 = 1, y_0 = 2$ with $h = 0.025$:

- $k_1 = \frac{1}{2} = 0.500000$
- $k_2 = f(1.0125, 2 + 0.5(0.0125)) = \frac{1.0125}{2.00625} \approx 0.504673$
- $k_3 = f(1.0125, 2 + 0.504673(0.0125)) = \frac{1.0125}{2.006308} \approx 0.504658$
- $k_4 = f(1.025, 2 + 0.504658(0.025)) = \frac{1.025}{2.012616} \approx 0.509287$

Resulting Step:

$$y(1.025) \approx 2 + \frac{0.025}{6}(0.5 + 2(0.504673) + 2(0.504658) + 0.509287)$$

$$y(1.025) \approx 2.006240$$

Effect of Step Size h

As expected accuracy improves significantly as h decreases:

Step Size (h)	Approx $y(2)$	Global Error
0.025	2.645751	$\sim 10^{-8}$
0.010	2.645751	$\sim 10^{-10}$
0.001	2.645751	$\sim 10^{-14}$

- The **Local Truncation Error** is $O(h^5)$.
- The **Global Truncation Error** is $O(h^4)$.
- RK4 tracks the exact solution much more closely than Euler or Heun methods.

Exercise

Write a Matlab script and solve the same initial value problem using the RK4 method.